Bihar Engineering University, Patna

B.Tech. 8th Semester Examination, 2024

Course: B.Tech. *Time: 03 Hours*

Code: 101827 *Subject: Construction Engineering Materials* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.

(ii) There are NINE questions in this paper.

(iii) Attempt FIVE questions in all.

(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / write the answer of the following (Any seven):*** **[2 x 7 = 14]**

(a) For complete hydration of cement the water cement ratio needed is
(i) 0.65 p (ii) 0.85 p (iii) 0.6 p (iv) 0.8 p

(b) Ultimate strength of cement is influenced by which one of the following?
(i) Tricalcium silicate (ii) Dicalcium silicate
(iii) Tricalcium aluminate (iv) Tetracalcium alumina-ferrite

(c) The approximate ratio of strength of 15cm x 30cm concrete cylinder to that of 15 cm cube of the same concrete is
(i) 1.25 (ii) 1.00 (iii) 0.85 (iv) 0.5

(d) The role of superplasticizer in a cement paste is to
(i) Disperse the particles
(ii) Disperse the particles and to remove air bubbles
(iii) Disperse the particles, remove air bubbles and to retard setting
(iv) Retard setting

(e) The minimum compressive strength of first class bricks should be
(i) 5 N/mm² (ii) 7.5 N/mm² (iii) 9 N/mm² (iv) 10 N/mm²

(f) The average compressive strength of a burnt clay brick is less than 12.5 N/mm². The allowable rating of efflorescence is
(i) Moderate (ii) Serious (iii) Heavy (iv) Zero

(g) What is value of portable water, as specified by IS 456-2000?
(i) Equal to 7 (ii) Between 6 and 9 (iii) Less than 6 (iv) Not less than 6

(h) Jumper is a tool used for
(i) Testing of stones (ii) Quarrying of stones
(iii) Dressing of stones (iv) None

(i) The ratio of young's modulus of high tensile steel to that of mild steel is about
(i) 0.5 (ii) 1.0 (iii) 1.5 (iv) 2.0

(j) How much SiO_2 is present in the glass which is drawn into fibres?
(i) 55% (ii) 15% (iii) 10% (iv) 4%

Q.2 (a) List the various laboratory tests for assessing the quality of cement and their importance. **[7]**

(b) What is slump test of concrete? How is it carried out? Mention recommended slumps for concrete for different purposes. **[7]**

Q.3 (a) Write about the following test on clay bricks and mention the desired test limits as per Indian standards **[7]**
(i) water absorption (ii) presence of soluble salts (iii) hardness (iv) soundness

(b) Explain various defects in bricks. **[7]**

Q.4 (a) What are the desirable qualities of a good building stones? Discuss them. **[7]**

(b) What is lime? Explain the classification of limes. **[7]**

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Q.5 (a) What are the purposes of heat treatment processes for steel? Mention the factor on which the heat treatment processes depends. [7]

(b) What are the properties of FRP? Mention the applications of FRP in building industry. [7]

Q.6 (a) Define mortar and discuss its varieties on the basis of nature of application [7]

(b) What is gypsum plaster? Mention its properties and uses. [7]

Q.7 (a) How are pre-cast concrete products prepared? Mention the advantages and disadvantages of pre-cast concrete. [7]

(b) What is carbon-carbon composite (CCC)? Write down the properties and application of carbon-carbon composites. [7]

Q.8 (a) How is construction and demolition waste recycled? [7]

(b) What is the problem with construction and demolition waste recycling? [7]

Q.9 Write Short notes on *any four* of the following [3.5 x 4=14]

(a) Durability

(b) Classification of bricks

(c) Ready- mix concrete

(d) Biodegradable plastic

(e) Nano composites

(f) 3D printing

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